

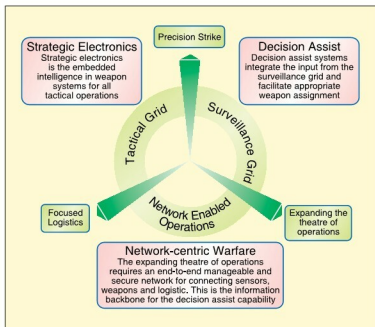
to the Ministry of Defence (MoD) for indigenous defence electronics production with some special licences.

Ghosh declares, "Over the next five years these licences will open a domestic addressable market of over Rs 200 billion for Tata Power SED, which includes upgrades of existing platforms."

The strategic electronics industry comprises lead system integrators (tier-I class) and component and sub-system manufacturers (tier-II class). System integration with outsourced components and sub-systems is the most viable business option in this segment. Components need huge production size but the chances of discouraging factors like volume rejection and technology obsolescence are limited. Sub-systems may require a good design and technology base.

Business model

After 1990, in the post-cold-war era, traditional defence business model has evolved to a great extent. Domestic business focused on platforms and exports at marginal cost and improved returns are no longer regular practices. The accelerated rate of change in civilian technology is also reflected in the military systems. More and more COTS



New operational model for strategic electronics (Courtesy: Tata Power, SED)

(commercial off-the-shelf) systems are replacing the old MIL (made-in-lab) systems. Nonetheless, associated with this shift are rising costs and complexities.

A well-known veteran from this field, Col. Kuber, vice president-aerospace and defence advisory service, Religare, points out, "Public-private partnership with mission-based procurement, joint development of system architecture, and integration of niche capability-based services and component suppliers may create a realistic approach to cope up with the changed business environment in the strategic electronics industry."

His comment is supported by the

success story of Axis Aerospace and Technologies. This premier supplier to ministry of Defence (MoD) follows a hybrid model of organic and inorganic growth. They nurture several boutique technology firms either through acquisition or partnership. This reduces their R&D investment to a great extent.

Moreover, adoption of a new approach towards the procurement of weapons and systems is significant. In the near future, the war-fighting-capability-driven approach can play a key role in the ecosystem rather

than the existing general staff qualitative requirement based one.

Value chain

The "create true value first and then create an illusion of even greater value" approach of the consumer electronics industry is not applicable to the strategic electronics industry. Here an entrepreneur is expected to support near-to-future technology at the same time. The return on investment is associated with an exceptionally high silent period. These contradictory situations put India in a unique position in the global value chain. India's legendary knowledge edge and its skilled, industrious manpower are right now upgrading it to high end as far as innovation is concerned. At the same time, the pace of development is not up to the mark due to high investment crunch.

In the value proposition term, a significant change has been observed in the last five years. Preference for domestic route over the import of technology is increasing. A steady shift towards 'develop and make' model is prominent from the previous 'buy' and 'buy-and-make' policy. As a result, the opportunities for the Indian talent pool are also increasing. Moreover, this new model has opened up opportunities for

"Defence electronics is a strategic industry—more than mere commerce. Issues of obsolescence, technology denial and restricted trade operate here. Moreover, it is monopoly—government is the market and the market maker."

—Sujoy Ghosh, project director, TATA Power, strategic electronics division

